

SOCI 3102 Social Networks and Social Capital

Problem Set 3: 10% of your score

Everyone will get a dataset about academic citation networks. Please find your assigned dataset in the Excel form by your name and corresponding dataset number. Next, please download your assigned dataset from the Blackboard and finalize your assignments.

Data Description

Each dataset contains all articles published by a university/college in 2011 and the research field of the article. More specifically, we have two columns in each dataset and the first column is the research field of published papers in that university; the second column is the research field of this paper's reference. Notice that every published paper can cite multiple other references.

For instance,

“Information flow through strong and weak ties in intraorganizational social networks” [focal paper], “Social Science” [focal paper’s field], “Medical Innovation: A Diffusion Study”[reference paper x], “Medicine”[reference paper x’s field]

Then, we will select “social science” as a starting point and “medicine” as an ending point in our dataset. Therefore, we could construct networks and observe some patterns.

Tasks

You need to construct the networks based on the dataset and complete the following tasks in RStudio,

- 1) Calculate the network density of your constructed networks. (1%)
- 2) Use random walk algorithm in R to detect your network communities and draw the clustered picture. (2%)
- 3) To make your pictures look better, please ensure different clusters have different colors and change the color of the original community to another. (3%)
- 4) Based on the previous analysis, please give your observations and interpretations about the pattern of the citation network of your assigned dataset (No more than 300 words). (4%)